
Chapter 6: Perimeter and Area

Worksheet 6: Perimeter of Plane Figures

Learning Objective: Students will calculate perimeters of triangles, quadrilaterals, circles (circumference), and composite figures; apply perimeter concepts to real-life problems.

Worked Example

Find the perimeter of a triangle with sides 8.5 cm, 6.3 cm, and 7.2 cm.

Solution: Perimeter = $8.5 + 6.3 + 7.2 = 22.0$ cm

Circumference of a circle: $C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm (for $r = 7$ cm).

Questions 1–5: Easy / Recall

Q1. Find the perimeter of a square with side 9 cm.

Q2. Find the perimeter of a rectangle: length 14 cm, breadth 9 cm.

Q3. A triangle has sides 5 cm, 12 cm, and 13 cm. Find its perimeter.

Q4. What is the circumference of a circle with radius 7 cm? (Use $\pi = \frac{22}{7}$)

Q5. A regular pentagon has each side 6 cm. Find its perimeter.

Questions 6–10: Basic Application

Q6. A rectangular park is 80 m long and 55 m wide. Find the length of fencing needed to go around it.

Q7. The circumference of a circle is 88 cm. Find its radius. (Use $\pi = \frac{22}{7}$)

Q8. A rhombus has each side 7.5 cm. Find its perimeter.

Q9. The perimeter of a rectangle is 84 cm. If its length is 25 cm, find its breadth.

Q10. A semicircle has diameter 14 cm. Find its perimeter (curved part + diameter). Use $\pi = \frac{22}{7}$.

Questions 11–15: Practice and Skill Building

Q11. An L-shaped path has outer dimensions 10 m $\times$ 8 m with a rectangular notch of 4 m $\times$ 3 m cut from one corner. Find the perimeter of the L-shape.

Q12. A farmer has 240 m of wire to fence a rectangular field. He wants the length to be twice the breadth. Find the dimensions.

Q13. Riya runs 4 laps around a circular track of radius 35 m. Find the total distance she runs. (Use $\pi = \frac{22}{7}$)

Q14. A square and a circle have the same perimeter of 88 cm. Which has the larger perimeter? Find both areas and compare. (Use $\pi = \frac{22}{7}$)

Q15. A regular hexagon has perimeter 72 cm. Find the radius of its circumscribed circle (equals side length).

Questions 16–18: Higher Order Thinking

Q16. A field is in the shape of a right triangle with legs 24 m and 10 m. Find the hypotenuse and then the perimeter. (Use Pythagoras.)

Q17. A rectangular garden has perimeter 60 m. The difference between length and breadth is 6 m. Find the length and breadth. What is the area?

Q18. Three circles, each of radius 4 cm, are placed touching each other. Find the perimeter of the triangular region formed between them. (Hint: it is an equilateral triangle with the sides reduced by arc lengths.)

Questions 19–20: Real-Life Applications

Q19. A school running track has two straight sections of 100 m each and two semicircular ends of radius 35 m. Find the total length of the track. (Use $\pi = \frac{22}{7}$)

- Q20.** A rectangular swimming pool ($50\text{ m} \times 25\text{ m}$) has a 2-m-wide path all around it. Find the perimeter of the outer edge of the path and the perimeter of the pool. What is the extra perimeter of the outer edge?

Reflection Question: Can two different shapes have the same perimeter but different areas? Draw examples to justify.